

# Proposed Outline Method For Decryption Of Kryptos K4

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May 2020

## Overview

This paper puts forward what is believed to be the underlying method necessary for deciphering the Kryptos K4 cipher. It is at the time of publishing believed to be an original piece of work that draws upon and extends the discoveries already put into the public domain by the author's previous papers, 'Valkyrie' in 2013 and 'The Machine' series of long and short form papers submitted to the Kryptos Yahoo group in 2017. The paper does make some attempts at the necessary puzzle steps themselves, but at its core there is some essential cryptanalysis that the more mathematically inclined may find useful in order to rule this in or out as viable framework within which the entire K4 section of Kryptos will eventually play out. Most importantly this method would appear to set parameters on what can be valid solutions. It will also show that a final contiguous 97 letter answer must be protected to some degree by 4 distinct puzzle steps that would appear to pivot around the underlying methodology outlined in this paper.

The overall premise is that K4 is segmented into 7 distinct X shaped layers that comprise 16 characters, broken down into 2 distinct groupings of 8 each representing one arm of a composite X shape. The 8 character groups have a distinct pattern that is built up 2 characters at a time. Each of the 2 characters are a palimpsest taken directly from the cipher text itself. The 2 characters are read by *encrypting* them against the relevant characters in the K4 cipher using the standard Quagmire III method and Kryptos alphabet. Two further methods are used to obfuscate this process. The first is a simple keyed Caesar shift that is applied after each successive block of 8 has been encrypted. The second method of obfuscation is an alignment process that makes use of the plaintext to indicate the parameters that determine the shape and form for the 8 block character groupings in the cipher text. Whilst this process sounds daunting, it is hoped that the visual breakdown given herein, will show that it is actually more consistent than it sounds from the description above.

## Limitations

At the time of writing, the complete alignment process is still a work in progress as it will be seen that this part of the decryption scheme represents the actual 'sluething' required to find a route through to the final decrypt. The proposed underlying method however generates all known solutions in their correct positions when combined with a brute force approach to fixing the positions, which then allows the alignment method to be conjectured retrospectively for each case.

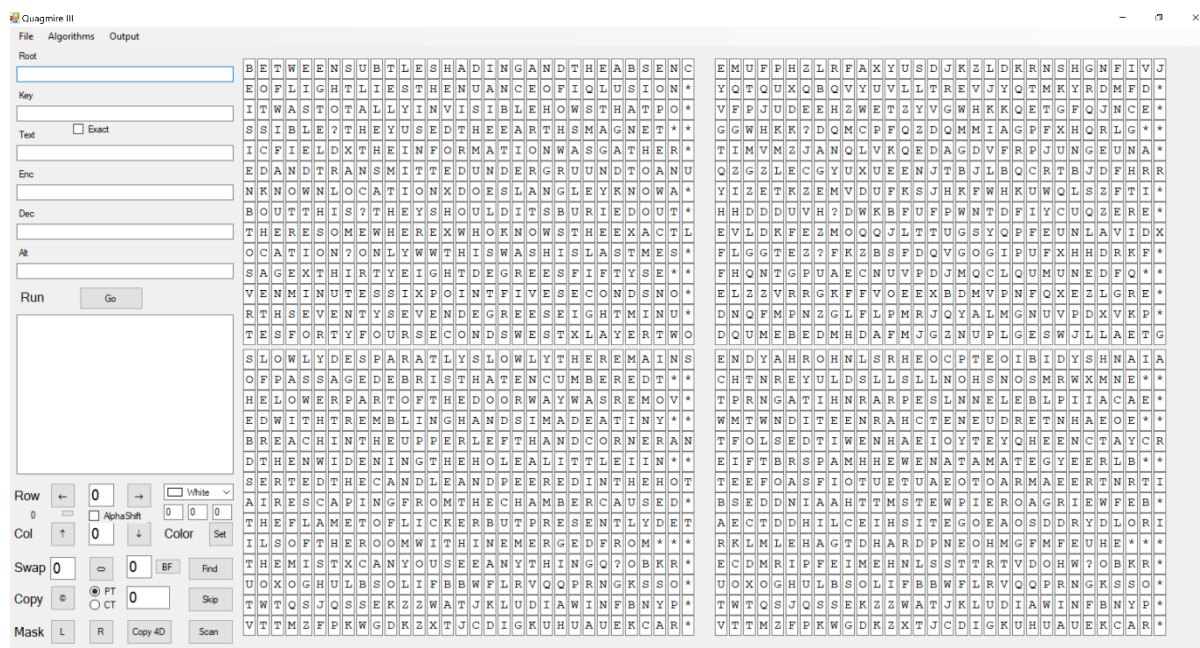
## Prerequisites

In order to successfully use the method described herein it is probably easiest to use a software tool that is capable of applying the following transforms to the Kryptos Cipher Text.

- A method of shifting individual rows to the right or left over a field of 32 characters, whereby a right shift will make the current 32<sup>nd</sup> character the 1<sup>st</sup> character and shift all subsequent character positions right by 1 position and so on.
- A method of transforming the cipher text in bulk by use of a keyed Caesar cipher where the key is KRYPTOS. This can in fact be done in by using the decryption tableau on the sculpture itself, but software is recommended for expediency.
- A method of showing the plaintext and cipher text side by side that allows transforms such as row shifts to be applied and shown simultaneously.

The discoveries shared in this paper make use of bespoke software written by the author in order to achieve these functional requirements, but written tables are provided in order to allow access to the same outputs as the software.

As can be seen here, the software gives a simultaneous view of both the plaintext and the cipher text. Any positional shifts applied to a given row are made in both views.



Additional features of the software also allow for easy pattern matching and the running of algorithms that can perform large scale scans of the PT and CT once compelling patterns have been identified. The most important recent addition has been the addition of a button that can cycle the entire PT and CT through a keyed Caesar shift one letter to the right each time.

## Method

It can be seen from the author's previous papers starting with 'Valkyrie' in 2013 that there is an interesting pattern formed in K1 when the first row is 'left shifted' 13 characters. The result of this transformation is shown here for the cipher text.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| U | S | D | J | K | Z | L | D | K | R | N | S | H | G | N | F | I | V | J | E | M | U | F | P | H | Z | L | R | F | A | X | Y |  |
| Y | Q | T | Q | U | X | Q | B | Q | V | Y | U | V | L | L | T | R | E | V | J | Y | Q | T | M | K | Y | R | D | M | F | D |   |  |
| V | F | P | J | U | D | E | E | H | Z | W | E | T | Z | Y | V | G | W | H | K | K | O | E | T | G | F | O | J | N | C | E |   |  |

Our initial clue of interest in the previous papers was that the characters UNYY when 'encrypted' using the Kryptos Quagmire III scheme with a keyword of OBKR gave a resulting string of Kryp. The most intriguing part of this transformation was what happened to the corresponding plaintext under the same conditions when a left shift of 13 was applied to the first row.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| S | H | A | D | I | N | G | A | N | D | T | H | E | A | B | S | E | N | C | B | E | T | W | E | E | N | S | U | B | T | L | E |  |
| E | O | F | L | I | G | H | T | L | I | E | S | T | H | E | N | U | A | N | C | E | O | F | I | Q | L | U | S | I | O | N |   |  |
| I | T | W | A | S | T | O | T | A | L | L | Y | I | N | V | I | S | I | B | L | E | H | O | W | S | T | H | A | T | P | O |   |  |

It was also noted that a 'right shift' of 13 when applied to the third row in the above arrangement results in the following pattern of note.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| U | S | D | J | K | Z | L | D | K | R | N | S | H | G | N | F | I | V | J | E | M | U | F | P | H | Z | L | R | F | A | X | Y |  |
| Y | Q | T | Q | U | X | Q | B | Q | V | Y | U | V | L | L | T | R | E | V | J | Y | Q | T | M | K | Y | R | D | M | F | D |   |  |
| K | K | Q | E | T | G | F | Q | J | N | C | E |   | V | F | P | J | U | D | E | H | Z | W | E | T | Z | Y | V | B | W | H |   |  |

The string UNYYCK when ‘encrypted’ against OBKRUE as the key gives KRYPTO. In the previous papers the reversal of direction for the characters K and C was explained away as an artefact of the embedded steganography that hides the coordinates (retained here as the blue and grey box shading), but more recent discoveries have allowed for a distinct separation of these two schemes, which most likely means that the machine aspect of the whole piece was something that should have only become apparent after K4 was broken.

In early 2020 an intriguing property of row 2 was found that had been continually overlooked, but now appeared to make sense of the above picture. The YY configuration in the second row is actually repeated consecutively meaning that a ‘left shift’ of 10 characters will still result in the same YY configuration needed to form UNYY in the left hand arrangement. If we leave the third row in its original state under this transformation then a new pattern becomes apparent.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| U | S | D | J | K | Z | L | D | K | R | N | S | H | G | N | F | I | V | J | E | M | U | F | P | H | Z | L | R | F | A | X | Y |  |
| Y | U | V | L | L | T | R | E | V | J | Y | Q | T | M | K | Y | R | D | M | F | D |   | Y | Q | T | Q | U | X | Q | B | Q | V |  |
| V | F | P | J | U | D | E | E | H | Z | W | E | T | Z | Y | V | G | W | H | K | K | Q | E | T | G | F | Q | J | N | C | E |   |  |

If the right hand side is read upside down then we have a new string UNYY CKBF which when ‘encrypted’ with OBKR UOXO becomes KRYPTOS L although this result will be revisited again later.

### Row alignment scheme

When looking at the necessary row shifts required for the above result it quickly becomes apparent that a sophisticated row alignment scheme is in play. After some initial experimentation it was reasoned that we might be expected to do something symmetrical in K4 corresponding to any row movements carried out in K1/K2. Doing the shift of 13 left in row 1 and 10 left in row 2 leads to the following set of results for the cipher text as we have already seen previously. *(It has not passed the authors attention that what appears to be a null cipher of ABC exists in the right hand column in partial highlight. This lead was followed up, but dismissed in favour of the following results).*

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| U | S | D | J | K | Z | L | D | K | R | N | S | H | G | N | F | I | V | J | E | M | U | F | P | H | Z | L | R | F | A | X | Y |  |
| Y | U | V | L | L | T | R | E | V | J | Y | Q | T | M | K | Y | R | D | M | F | D |   | Y | Q | T | Q | U | X | Q | B | Q | V |  |
| V | F | P | J | U | D | E | E | H | Z | W | E | T | Z | Y | V | G | W | H | K | K | Q | E | T | G | F | Q | J | N | C | E |   |  |

If we place the first 3 rows of K1 above the 3 full rows of K4 and shift rows symmetrically so that a left shift in row one equates to a right shift in row twenty eight and a right shift in row twenty seven equates to a left shift in row two etc, then we see the following pattern for the 13 and 10 left shift applied above.

Symmetrically applying a right shift of 13 to row 28 and a right shift of 10 to row 27 in K4 yields the following result.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| U | O | X | O | G | H | U | L | B | S | O | L | I | F | B | B | W | F | L | R | V | Q | Q | P | R | N | G | K | S | S | O | * |
| A | W | I | N | F | B | N | Y | P |   | T | W | T | Q | S | J | Q | S | S | E | K | Z | Z | W | A | T | J | K | L | U | D | I |
| G | K | U | H | U | A | U | E | K | C | A | R |   | V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J | C | D | I |

The plain text in rows 1-3 under the same shift transformation gives.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| S | H | A | D | I | N | G | A | N | D | T | H | E | A | B | S | E | N | C | B | E | T | W | E | E | N | S | U | B | T | L | E |  |
| E | S | T | H | E | N | U | A | N | C | E | O | F | I | Q | L | U | S | I | O | N |   | E | O | F | L | I | G | H | T | L | I |  |
| I | T | W | A | S | T | O | T | A | L | L | Y | I | N | V | I | S | I | B | L | E | H | O | W | S | T | H | A | T | P | O |   |  |

Many parts of what appears to be the cipher are actually hints rather than explicit end results, so separating the hints from meaningful actionable results can sometimes be a tortuous process, but it is suspected that the puzzle setter is purposely drawing us through a defined route in order to create a crib in order to solve the remainder of the puzzle. The above hint is no exception as it appears to correlate perfectly with the clue ID BY ROWS. Looking for further interesting properties of this arrangement led to a range of additional observations.

### The Mist X

A new algorithm was trialed in order to spot any symmetrical correlations between the first three rows of Kryptos in K1/K2 and the bottom three rows in K4. The general goal of the algorithm was to identify groupings of 3 letters that shared similar spatial arrangements in the first 3 rows of K1/2 and the full three rows of K4. The yield from such an algorithm was expected to be low given that visual inspection had not provided any other example other than IAC that was initially found intriguing due to both the specific letters involved as well as the output string that was being derived. It was therefore surprising when the algorithm turned up over 2000 correlations, so it was reasoned that the alignment scheme must be being driven by a more restrictive process. This led to a tightening of the algorithm to find only pairs that made a corresponding X pattern when overlaid. This reduced the yield to about 60 candidates and some of these were explored. Amazingly one such pair had been present all along in the alignment scheme already detailed and was simply overlooked due to cognitive bias based on the belief that the 3 letter marker would likely be CIA.

The most obvious feature of this arrangement as highlighted previously is the double DI block at the right hand end of K4. Looking closer it would appear that something was missed previously, which is a symmetric correlation in K1/2 and K4 as shown below.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| S | H | A | D | I | N | G | A | N | D | T | H | E | A | B | S | E | N | C | B | E | T | W | E | E | N | S | U | B | T | L | E |   |
| E | S | T | H | E | N | U | A | N | C | E | O | F | I | Q | L | U | S | I | O | N | * | E | O | F | L | I | G | H | T | L | I |   |
| I | T | W | A | S | T | O | T | A | L | L | Y | I | N | V | I | S | I | B | L | E | H | O | W | S | T | H | A | T | P | O | * |   |
| S | S | I | B | L | E | ? | T | H | E | Y | U | S | E | D | T | H | E | E | A | R | T | H | S | M | A | G | N | E | T | * | * |   |
| I | C | F | I | E | L | D | X | T | H | E | I | N | F | O | R | M | A | T | I | O | N | W | A | S | G | A | T | H | E | R | * |   |
| E | D | A | N | D | T | R | A | N | S | M | I | T | T | E | D | U | N | D | E | R | G | R | U | U | N | D | T | O | A | N | U |   |
| N | K | N | O | W | N | L | O | C | A | T | I | O | N | X | D | O | E | S | L | A | N | G | L | E | Y | K | N | O | W | A | * |   |
| B | O | U | T | T | H | I | S | ? | T | H | E | Y | S | H | O | U | L | D | I | T | S | B | U | R | I | E | D | O | U | T | * |   |
| T | H | E | R | E | S | O | M | E | W | H | E | R | E | X | W | H | O | K | N | O | W | S | T | H | E | E | X | A | C | T | L |   |
| O | C | A | T | I | O | N | ? | O | N | L | Y | W | W | T | H | I | S | W | A | S | H | I | S | L | A | S | T | M | E | S | * |   |
| S | A | G | E | X | T | H | I | R | T | Y | E | I | G | H | T | D | E | G | R | E | E | S | F | I | F | T | Y | S | E | * | * |   |
| V | E | N | M | I | N | U | T | E | S | S | I | X | P | O | I | N | T | F | I | V | E | S | E | C | O | N | D | S | N | O | * |   |
| R | T | H | S | E | V | E | N | T | Y | S | E | V | E | N | D | E | G | R | E | E | S | E | I | G | H | T | M | I | N | U | * |   |
| T | E | S | F | O | R | T | Y | F | O | U | R | S | E | C | O | N | D | S | W | E | S | T | I | D | B | Y | R | O | W | S | * |   |
|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| S | L | O | W | L | Y | D | E | S | P | A | R | A | T | L | Y | S | L | O | W | L | Y | T | H | E | R | E | M | A | I | N | S |   |
| O | F | P | A | S | S | A | G | E | D | E | B | R | I | S | T | H | A | T | E | N | C | U | M | B | E | R | E | D | T | * | * |   |
| H | E | L | O | W | E | R | P | A | R | T | O | F | T | H | E | D | O | O | R | W | A | Y | W | A | S | R | E | M | O | V | * |   |
| E | D | W | I | T | H | T | R | E | M | B | L | I | N | G | H | A | N | D | S | I | M | A | D | E | A | T | I | N | Y | * | * |   |
| B | R | E | A | C | H | I | N | T | H | E | U | P | P | E | R | L | E | F | T | H | A | N | D | C | O | R | N | E | R | A | N |   |
| D | T | H | E | N | W | I | D | E | N | I | N | G | T | H | E | H | O | L | E | A | L | I | T | T | L | E | I | I | N | * | * |   |
| S | E | R | T | E | D | T | H | E | C | A | N | D | L | E | A | N | D | P | E | E | R | E | D | I | N | T | H | E | H | O | T |   |
| A | I | R | E | S | C | A | P | I | N | G | F | R | O | M | T | H | E | C | H | A | M | B | E | R | C | A | U | S | E | D | * |   |
| T | H | E | F | L | A | M | E | T | O | F | L | I | C | K | E | R | B | U | T | P | R | E | S | E | N | T | L | Y | D | E | T | A |
| I | L | S | O | F | T | H | E | R | O | O | M | W | I | T | H | I | N | E | M | E | R | G | E | D | F | R | O | M | * | * | * |   |

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| T | H | E | M | I | S | T | X | C | A | N | Y | O | U | S | E | E | A | N | Y | T | H | I | N | G | Q | ? | O | B | K | R | * |  |
| U | O | X | O | G | H | U | L | B | S | O | L | I | F | B | B | W | F | L | R | V | N | O | R | T | H | E | A | S | T | O | * |  |
| A | W | I | N | F | B | N | Y | P | * | T | W | T | Q | S | J | Q | S | S | E | K | Z | Z | W | A | T | J | K | L | U | D | I |  |
| G | K | U | H | U | A | U | E | K | C | A | R | * | V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J | C | D | I |  |

This was the exact symmetry that the algorithm had been designed to spot, so it was pleasing to find that it had been initially overlooked in the initial results for the KRYPTOS L alignment.

A number of discoveries then appeared to cascade from this simple rearrangement.

The first was that the double DI block is in actual diametric opposition to the string DI in the rearranged K1 when the blocks are connected diagonally. It was whilst examining this relationship that it became apparent another clue had inadvertently been stumbled across.

If we continue the diagonal climb of ANL to the twenty fifth row then we are presented with the following letters of note in the rows for K4.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| T | H | E | M | I | S | T | X | C | A | N | Y | O | U | S | E | E | A | N | Y | T | H | I | N | G | Q | ? | O | B | K | R | * |  |
| U | O | X | O | G | H | U | L | B | S | O | L | I | F | B | B | W | F | L | R | V | N | O | R | T | H | E | A | S | T | O | * |  |
| A | W | I | N | F | B | N | Y | P | * | T | W | T | Q | S | J | Q | S | S | E | K | Z | Z | W | A | T | J | K | L | U | D | I |  |
| G | K | U | H | U | A | U | E | K | C | A | R | * | V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J | C | D | I |  |

It's reasoned that this is the fabled 'CANDLE' clue that many have conjectured to be significant in any potential decrypt, although it can be seen that this is an anagrammed homonym of CANDIL. The relationships that now start to fall out of these observations are profound. The first is that if we continue the diagonal climb upwards whilst simultaneously climbing down for K1/2 then the two lines make the intersection shown below.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| S | H | A | D | I | N | G | A | N | D | T | H | E | A | B | S | E | N | C | B | E | T | W | E | E | N | S | U | B | T | L | E |   |   |
| E | S | T | H | E | N | U | A | N | C | E | O | F | I | Q | L | U | S | I | O | N | * | E | O | F | L | I | G | H | T | L | I |   |   |
| I | T | W | A | S | T | O | T | A | L | E | Y | I | N | V | I | S | I | B | L | E | H | O | W | S | T | H | A | T | P | O | * |   |   |
| S | S | I | B | L | E | ? | T | H | E | Y | U | S | E | D | T | H | E | E | A | R | T | H | S | M | A | G | N | E | T | * | * |   |   |
| I | C | F | I | E | L | D | X | T | H | E | I | N | F | O | R | M | A | T | I | O | N | W | A | S | G | A | T | H | E | R | * |   |   |
| E | D | A | N | D | T | R | A | N | S | M | I | T | T | E | D | U | N | D | E | R | G | R | U | U | N | D | T | O | A | N | U |   |   |
| N | K | N | O | W | N | L | O | C | A | T | I | O | N | X | D | O | E | S | L | A | N | G | L | E | Y | K | N | O | W | A | * |   |   |
| B | O | U | T | T | H | I | S | ? | T | H | E | Y | S | H | O | U | L | D | I | T | S | B | U | R | I | E | D | O | U | T | * |   |   |
| T | H | E | R | E | S | O | M | E | W | H | E | R | E | X | W | H | O | K | N | O | W | S | T | H | E | E | X | A | C | T | L |   |   |
| O | C | A | T | I | O | N | ? | O | N | L | Y | W | W | T | H | I | S | W | A | S | H | I | S | L | A | S | T | M | E | S | * |   |   |
| S | A | G | E | X | T | H | I | R | T | Y | E | I | G | H | T | D | E | G | R | E | E | S | F | I | F | T | Y | S | E | * | * |   |   |
| V | E | N | M | I | N | U | T | E | S | S | I | X | P | O | I | N | T | F | I | V | E | S | E | C | O | N | D | S | N | O | * |   |   |
| R | T | H | S | E | V | E | N | T | Y | S | E | V | E | N | D | E | G | R | E | E | S | E | I | G | H | T | M | I | N | U | * |   |   |
| T | E | S | F | O | R | T | Y | F | O | U | R | S | E | C | O | N | D | S | W | E | S | T | I | D | B | Y | R | O | W | S | * |   |   |
| S | L | O | W | L | Y | D | E | S | P | A | R | A | T | L | Y | S | L | O | W | L | Y | T | H | E | R | E | M | A | I | N | S |   |   |
| O | F | P | A | S | S | A | G | E | D | E | B | R | I | S | T | H | A | T | E | N | C | U | M | B | E | R | E | D | T | * | * |   |   |
| H | E | L | O | W | E | R | P | A | R | T | O | F | T | H | E | D | O | O | R | W | A | Y | W | A | S | R | E | M | O | V | * |   |   |
| E | D | W | I | T | H | T | R | E | M | B | L | I | N | G | H | A | N | D | S | I | M | A | D | E | A | T | I | N | Y | * | * |   |   |
| B | R | E | A | C | H | I | N | T | H | E | U | P | P | E | R | L | E | F | T | H | A | N | D | C | O | R | N | E | R | A | N |   |   |
| D | T | H | E | N | W | I | D | E | N | I | N | G | T | H | E | H | O | L | E | A | L | I | T | T | L | E | I | I | N | * | * |   |   |
| S | E | R | T | E | D | T | H | E | C | A | N | D | L | E | A | N | D | P | E | E | R | E | D | I | N | T | H | E | H | O | T |   |   |
| A | I | R | E | S | C | A | P | I | N | G | F | R | O | M | T | H | E | C | H | A | M | B | E | R | C | A | U | S | E | D | * |   |   |
| T | H | E | F | L | A | M | E | T | O | F | L | I | C | K | E | R | B | U | T | P | R | E | S | E | N | T | U | L | Y | D | E | T | A |
| I | L | S | O | F | T | H | E | R | O | O | M | W | I | T | H | I | N | E | M | E | R | G | E | D | F | R | O | M | * | * | * |   |   |
| T | H | E | M | I | S | T | X | C | A | N | Y | O | U | S | E | E | A | N | Y | T | H | I | N | G | Q | ? | O | B | K | R | * |   |   |
| U | O | X | O | G | H | U | L | B | S | O | L | I | F | B | B | W | F | L | R | V | N | O | R | T | H | E | A | S | T | O | * |   |   |
| A | W | I | N | F | B | N | Y | P | * | T | W | T | Q | S | J | Q | S | S | E | K | Z | Z | W | A | T | J | K | L | U | D | I |   |   |
| G | K | U | H | U | A | U | E | K | C | A | R | * | V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J | C | D | I |   |   |

Before we pursue this line of inquiry further it is important to make another observation, which is that there appears to be a direct correlation between this scheme and key target letters within the cipher text as shown by the following relationship. Remember the rows of the cipher text are in the same shift configuration as the plain text.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| S | H | A | D | I | N | G | A | N | D | T | H | E | A | B | S | E | N | C | B | E | T | W | E | E | N | S | U | B | T | L | E |  |
| E | S | T | H | E | N | U | A | N | C | E | O | F | I | Q | L | U | S | I | O | N | * | E | O | F | L | I | G | H | T | L | I |  |
| I | T | W | A | S | T | O | T | A | L | L | Y | I | N | V | I | S | I | B | L | E | H | O | W | S | T | H | A | T | P | O | * |  |

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| U | O | X | O | G | H | U | L | B | S | O | L | I | F | B | B | W | F | L | R | V | N | O | R | T | H | E | A | S | T | O | * |  |
| A | W | I | N | F | B | N | Y | P | * | T | W | T | Q | S | J | Q | S | S | E | K | Z | Z | W | A | T | J | K | L | U | D | I |  |
| G | K | U | H | U | A | U | E | K | C | A | R | * | V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J | C | D | I |  |

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| U | S | D | J | K | Z | L | D | K | R | N | S | H | G | N | F | I | V | J | E | M | U | F | P | H | Z | L | R | F | A | X | Y |  |
| Y | U | V | L | L | T | R | E | V | J | Y | Q | T | M | K | Y | R | D | M | F | D |   | Y | Q | T | Q | U | X | Q | B | Q | V |  |
| V | F | P | J | U | D | E | E | H | Z | W | E | T | Z | Y | V | G | W | H | K | K | Q | E | T | G | F | Q | J | N | C | E |   |  |

Encrypting UNYY CKBD against OBRKR UOXO gives Kryp TOSI. It is reasoned that KRYPTOS is a sensible first word of the decrypt in order to give the solver feedback that they are on the right decryption path, so I shall refer to this solve as crib number one as it is reasoned that it is trying to gently educate us as to what we might be expected to do as the puzzle progresses.

It should be pointed out at this stage that the evidence is effectively being presented in reverse order due to the manner in which the biggest break was acquired, so these initial puzzle solving steps actually came out of the cryptanalysis that centred round the known clues. I make this point as the second crib offers a more logical path to the next method of concealment than the lucky break that was initially utilised based on a hunch about the extraneous L in the decryption tableau. Once we see the method of concealment that drops out of the cryptanalysis then we are finally at liberty to start uncovering the row alignments needed for the plain text clues given out since 2010.

Before coming back to what is thought to be the second crib it will be useful to appraise the results of the cryptanalysis first, so that we have all parts of the decryption process to hand before proceeding further.

### Cryptanalysis of the known plain text

The results from the first crib were deemed intriguing enough to take seriously as a potential break, but this led to something of an impasse that persisted for a considerable period of time. Inspiration struck however when the tableau portion of the sculpture was considered. If the tableau is considered on a line by line basis then it effectively becomes the lookup table necessary for a keyed Caesar cipher. The key used of course is KRYPTOS and the extraneous L on the right of the tableau hints that we should consider a single 'right shift'. As a result of this idea the bespoke software in use was amended to shift all cipher and plain text right one character within the keyed Caesar scheme.

The result of this transformation on the first three (non-shifted) rows of the cipher text is as follows.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| F | N | V | G | T | I | K | M | Y | G | B | Z | P | V | A | E | L | R | K | M | E | R | Y | Q | A | I | H | Q | G | J | W | L |  |
| P | U | O | U | V | Z | U | C | U | W | P | V | W | M | M | O | Y | F | W | L | P | U | O | N | R | P | Y | E | N | G | E |   |  |
| W | G | T | L | V | E | F | F | I | K | X | F | O | K | P | W | H | X | I | R | R | U | F | O | H | G | U | L | Q | D | F |   |  |

After some guessing at what the next word might be following the letter L, the following arrangement was settled upon. ***This particular arrangement is now suspected to be incorrect as it was in use before the row alignment scheme was discovered, but rather usefully it did offer something of a boot strapping framework within which to explore letter ordering schemes, so these next 8 letters should be considered simply as a scaffolding framework as it will correspond to the later sense checks made against the known clues.***

From the standard position the row shifts applied here are a 'right shift' of one in row one, a 'left shift' of eight in row two and finally a 'left shift' of one in row three.

The result of this transformation is as follows:

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| L | F | N | V | G | T | I | K | M | Y | G | B | Z | P | V | A | E | L | R | K | M | E | R | Y | Q | A | I | H | Q | G | J | W |  |
| U | W | P | V | W | M | M | O | Y | F | W | L | P | U | O | N | R | P | Y | E | N | G | E |   | P | U | O | U | V | Z | U | C |  |
| G | T | L | V | E | F | F | I | K | X | F | O | K | P | W | H | X | I | R | R | U | F | O | H | G | U | L | Q | D | F |   | W |  |

The overall results from both this example and the known plaintext for K4 would imply that this configuration should be read off as MUFG and PKUE, which would imply something of a reversal of the previous method for reading off the intended letters. 'Encrypted' against the K4 string of GHUL BSOL this give OCATE SKY. This gives a combined decrypt for the first three rows as 'KRYPTOS LOCATE SKY'. Reviewing this paper prior to publishing into the forums it is somewhat obvious that my own personal bias was in play here due to my previous work on 'The machine' papers that argue the case for Kryptos as a whole being an analogue astronomical calculator. The letter selections will be amended later as the evidence base changes.

Something that immediately stands out is that the general format of the pattern when superimposed on the previous one forms something akin to an 'X' shape. Given that the pattern is built up two characters at a time it would appear to make sense of the statement 'X Layer Two'.

Pursuing this pattern further it was found that extending this method across consecutive rows of three moving downwards through the sculpture that it was capable of generating all known plaintext in the correct letter positions (apart from the I of BERLIN, more of which will be mentioned later).

In order to make sense of the method I shall refer to the X layer as a combination of the 'left hand' \ configuration and the 'right hand' configuration /. The results above give KRYPTOS L for the \ result and OCATE SKY for the / result.

A quick look up chart is now provided to show how such a scheme overlays onto K4 so that we can discern how to check this method against the known plain text. The \ configuration will be coloured in red and the / configuration blue. The colour convention is chosen as being red and blue in order to reflect the idea of North and South magnetic poles.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|--|
| E | C | D | M | R | I | P | F | E | I | M | E | H | N | L | S | S | T | T | R | T | V | D | O | H | W | ? | O | B | K | R |  |  |
| U | O | X | O | G | H | U | L | B | S | O | L | I | F | B | B | W | F | L | R | V | Q | Q | P | R | N | G | K | S | S | O |  |  |
| T | W | T | Q | S | J | Q | S | S | E | K | Z | Z | W | A | T | J | K | L | U | D | I | A | W | I | N | F | B | N | Y | P |  |  |
| V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J | C | D | I | G | K | U | H | U | A | U | E | K | C | A | R |  |  |

Remember that each time there is a change from the \ to the / configuration or move layer that we need to transform all cipher text under analysis by a right shift of one within the keyed Caesar scheme. I shall show this in more detail so that this step is explicit before proceeding.

Keyed Caesar alphabet: KRYPTOSABCDEFGHIJLMNQUVWXZ

Example text: CIPHER TEXT

Output from a 'right shift' of one: DJTIY OFZO.

It can be seen that the overall scheme is a palimpsest that reads characters directly off of the cipher 8 characters at a time in a composite X configuration. Each X layer exists within its own discrete set of three rows proceeding downwards through K1, K2 and K3. There are 7 full X layers and a final single letter to complete the full 97 run of K4.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| E | M | U | F | P | H | Z | L | R | F | A | X | Y | U | S | D | J | K | Z | L | D | K | R | N | S | H | G | N | F | I | V | J | 1 |
| Y | Q | T | Q | U | X | Q | B | Q | V | Y | U | V | L | L | T | R | E | V | J | Y | Q | T | M | K | Y | R | D | M | F | D |   |   |
| V | F | P | J | U | D | E | E | H | Z | W | E | T | Z | Y | V | G | W | H | K | K | Q | E | T | G | F | Q | J | N | C | E |   |   |
| G | G | W | H | K | K | ? | D | Q | M | C | P | F | Q | Z | D | Q | M | M | I | A | G | P | F | X | H | Q | R | L | G |   | 2 |   |
| T | I | M | V | M | Z | J | A | N | Q | L | V | K | Q | E | D | A | G | D | V | F | R | P | J | U | N | G | E | U | N | A |   |   |
| Q | Z | G | Z | L | E | C | G | Y | U | X | U | E | E | N | J | T | B | J | L | B | Q | C | R | T | B | J | D | F | H | R | R |   |
| Y | I | Z | E | T | K | Z | E | M | V | D | U | F | K | S | J | H | K | F | W | H | K | U | W | Q | L | S | Z | F | T | I | 3 |   |
| H | H | D | D | D | U | V | H | ? | D | W | K | B | F | U | F | P | W | N | T | D | F | I | Y | C | U | Q | Z | E | R | E |   |   |
| E | V | L | D | K | F | E | Z | M | O | Q | Q | J | L | T | T | U | G | S | Y | Q | P | F | E | U | N | L | A | V | I | D | X |   |
| F | L | G | G | T | E | Z | ? | F | K | Z | B | S | F | D | Q | V | G | O | G | I | P | U | F | X | H | H | D | R | K | F | 4 |   |
| F | H | Q | N | T | G | P | U | A | E | C | N | U | V | P | D | J | M | Q | C | L | Q | U | M | U | N | E | D | F | Q |   |   |   |
| E | L | Z | Z | V | R | R | G | K | F | F | V | O | E | E | X | B | D | M | V | P | N | F | Q | X | E | Z | L | G | R | E |   |   |
| D | N | Q | F | M | P | N | Z | G | L | F | L | P | M | R | J | Q | Y | A | L | M | G | N | U | V | P | D | X | V | K | P | 5 |   |
| D | Q | U | M | E | B | E | D | M | H | D | A | F | M | J | G | Z | N | U | P | L | G | E | W | J | L | L | A | E | T | G |   |   |
|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| E | N | D | Y | A | H | R | O | H | N | L | S | R | H | E | O | C | P | T | E | O | I | B | I | D | Y | S | H | N | A | I | A |   |
| C | H | T | N | R | E | Y | U | L | D | S | L | L | S | L | L | N | O | H | S | N | O | S | M | R | W | X | M | N | E |   | 6 |   |
| T | P | R | N | G | A | T | I | H | N | R | A | R | P | E | S | L | N | N | E | L | E | B | L | P | I | I | A | C | A | E |   |   |
| W | M | T | W | N | D | I | T | E | E | N | R | A | H | C | T | E | N | E | U | D | R | E | T | N | H | A | E | O | E |   |   |   |
| T | F | O | L | S | E | D | T | I | W | E | N | H | A | E | I | O | Y | T | E | Y | Q | H | E | E | N | C | T | A | Y | C | R | 7 |
| E | I | F | T | B | R | S | P | A | M | H | H | E | W | E | N | A | T | A | M | A | T | E | G | Y | E | E | R | L | B |   |   |   |
| T | E | E | F | O | A | S | F | I | O | T | U | E | T | U | A | E | O | T | O | A | R | M | A | E | E | R | T | N | R | T | I |   |
| B | S | E | D | D | N | I | A | A | H | T | T | M | S | T | E | W | P | I | E | R | O | A | G | R | I | E | W | F | E | B |   |   |
| A | E | C | T | D | D | H | I | L | C | E | I | H | S | I | T | E | G | O | E | A | O | S | D | D | R | Y | D | L | O | R | I | T |
| R | K | L | M | L | E | H | A | G | T | D | H | A | R | D | P | N | E | O | H | M | G | F | M | F | E | U | H | E |   |   |   |   |
| E | C | D | M | R | I | P | F | E | I | M | E | H | N | L | S | S | T | T | R | T | V | D | O | H | W | ? | O | B | K | R |   |   |
| U | O | X | O | G | H | U | L | B | S | O | L | I | F | B | B | W | F | L | R | V | Q | Q | P | R | N | G | K | S | S | O |   |   |
| T | W | T | Q | S | J | Q | S | S | E | K | Z | Z | W | A | T | J | K | L | U | D | I | A | W | I | N | F | B | N | Y | P |   |   |
| V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J | C | D | I | G | K | U | H | U | A | U | E | K | C | A | R |   |   |

Therefore the right shifts that will be applied to each X layer for our verification run are as follows.

| X Layer | \ Shift | / Shift | Layer contains      |
|---------|---------|---------|---------------------|
| 1       | 0       | 1       | KRYPTOSL OCATESKY   |
| 2       | 2       | 3       | XXXXXXXXX XNORTHEA  |
| 3       | 4       | 5       | STXXXXXXXX XXXXXXXX |
| 4       | 6       | 7       | XXXXXXXXX XXXXXXXXB |
| 5       | 8       | 9       | ERLENCLO CKXXXXXX   |
| 6       | 10      | 11      | XXXXXXXXX XXXXXXXX  |
| 7       | 12      | 13      | XXXXXXXXX XXXXXXXX  |

There is one interesting anomaly that is found during the initial verification run. This is that BERLIN actually resolves to BERLEN in positions 64-69. All other known plaintext is reproduced consistently in the correct positions using a defined method. As a result of this it is reasoned that the act of giving BERLIN as a public clue was actually an example of a duress code, whereby the information is seemingly equivalent to the real answer, but actually protects the underlying scheme from being reverse engineered.

For those who don't have the computational resources or time to compute the relevant shifts for each layer these are now provided in the following table.

| X Layer   Shift | Keyed Caesar Shifted Cipher Text                                                                         |
|-----------------|----------------------------------------------------------------------------------------------------------|
| 1 \0            | EMUFPHZLRFAXYUSDJKZLDRNSHGNFIVJ<br>YQTQUXQBQVYUUVLLTREVJYQTMKYRDMFD<br>VFPJUDEEHZWETZYVGWHKKQETGFQJNCE   |
| 1 /1            | FNVGTIKMYGBZPVAELRKMERYQAIHQGJWL<br>PUOUVZUCUWFPVWMMOYFWLPUONRPFYENGE<br>WGTLVEFFIKXFOKPWHXIRRUFOHGULQDF |
| 2 \2            | IIZJYY?FVQEOHVRVFVQQLCIOHKJVPNI<br>SLQXQRMUCUVNXVVGFCIFXHPOMWUIGWUC<br>VRIRNGEITWKWGGUMSDMNDVEPSDMFHJPP  |
| 2 /3            | JJKLPP?GWUFSIWYGWUUMDJSIRLWTQJ<br>AMUZUYNDVWQZPWHGDJGZITSNXVJHXVD<br>WYJYQHFJOXRXHHVNAENQEWFTAENGILTT    |
| 3 \4            | SNPIBTPIVKHZJTDQMTJRM TZRXUDPJBN<br>MMHHH ZKM?HRTFJZJARWBHJNSGZXPIOI<br>IKUHTJIPVCXXQUBBZLDSXAJIZWUEKNHY |
| 3 /5            | AQ TJCOTJWRIKLOEUNOLYNOKYZVETLCQ<br>NNII IKRN?IYOGLKLBXYCILQAHKZTJSJ<br>JRVIOLJTDWZ ZUVCKMEAZBLJKXVFRQIP |
| 4 \6            | MWNNDLO?MSOHFMJKYNENUCRMTQQJASM<br>MQKZDNCRGLIZRYCJVXKIWKRXRZLJMK<br>LWOOYAANSMMYELLTHJXYCZMKTLOWNAL     |
| 4 /7            | NXQQEMS?NASIGNLRPQFQVDYNOULBAN<br>NURKEQDYHMKYPDLWZRJXRYZYKMLNR<br>MXSSPBBQANNPFMMOILZPDKNROMSXQBM       |
| 5 \8            | MRYQKERAUZQZEKCYDIZKURPTEMSTBE<br>MYPKNJNMKVMIQKXUARPEZUNHOXZZINFU<br>NRMDIVCGVRZHCVNGLEFNGWJWMDHVRIWI   |
| 5 /9            | NYPURFYBVKUKFRDZPEJKRVYTOFNAOCF<br>NPTRQLQNRWNJURZVBYTFKVQISZKKJQGV<br>QYNEJWDHWYKIDWQHMFQGHXLXNEIWYJXJ  |
| 6 \10           | NXHPEUFORQJRRJRRPIXJPIJYEABYPU<br>HGEFWLHZXPELEGUJRPPURUMRGZZLNLU<br>AYHAPQZHUUPELXNHUPUOQEUHPXLUIU      |
| 6 /11           | QZITFVGSYULYYLYYTJZLTJLPFBCPTV<br>IHFTXMIKZTFMFHVLYTTVYVNYHKKMQMV<br>BPIBTUKIVVTFMZQIVTVSUFVITZMVJV      |
| 7 \12           | JXLPMWVJRCWOKNWRLHJWHSKWWOUJNHUG<br>WRXJQGMINTKKWCWONJNTNJWZHHWGPQ<br>JWWXLNMXRLJAWJANWLJLNGTNWWGJOGJR   |
| 7 /13           | LZMTNXWLYDXSRQXYMILXIARXXSVLQIVH<br>XYZLUHNJQORRXDXSQLQOQLXKIXXHTU<br>LXXZMQNZYMLBXLBQXMLMQHOQXXHLSHLY   |

So using the shift information from the tables above lets verify our first piece of known plain text at position 26. According to our lookup table we expect to find \_NORTHEA in the / configuration within the second row. The corresponding keyed Caesar shift for this block will be 3.

As we are dealing with 8 letters at a time we would expect to have a resulting string of \_ZEXPUXA that would encrypt with \_QQPRNGK to give \_NORTHEA.

Remember that we have a prototype form for the / configuration given by the previous example in the first X Layer:

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| L | F | N | V | G | T | I | K | M | Y | G | B | Z | P | V | A | E | L | R | K | M | E | R | Y | Q | A | I | H | Q | G | J | W |
| U | W | P | V | W | M | M | O | Y | F | W | L | P | U | O | N | R | P | Y | E | N | G | E |   | P | U | O | U | V | Z | U | C |
| G | T | L | V | E | F | F | I | K | X | F | O | K | P | W | H | X | I | R | R | U | F | O | H | G | U | L | Q | D | F |   | W |

Let's examine the rows without any shifts applied to see if the requisite letters can be found in the expected rows. We would expect the Z to be in row 2 and the EX to be in row 3. We would expect them to present as Z\_ and XE, so that they can be read according to the previously discovered configuration rules as \_ZEX. Similarly we would expect to find PU in row 1 and XA in row 2, read in the order PUXA.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| J | J | K | L | P | P | ? | G | W | U | F | S | I | W | Y | G | W | U | U | M | D | J | S | I | R | L | W | T | Q | J |   |   |  |
| A | M | U | Z | U | Y | N | D | V | W | Q | Z | P | W | H | G | D | J | G | Z | I | T | S | N | X | V | J | H | X | V | D |   |  |
| W | Y | J | Y | Q | H | F | J | O | X | R | X | H | H | V | N | A | E | N | Q | E | W | F | T | A | E | N | G | I | L | T | T |  |

This block is found to pass muster in terms of the requirements, so let's now try and arrange it as closely as possible to the previously found / configuration. Remember that this is a first pass verification step to make sure the requisite letters are present, so the distance between the two sides and their relative separation distance remains arbitrary at this point until the last part of the deciphering scheme is discussed.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| U | M | D | J | S | I | R | L | W | T | Q | J |   |   | J | J | K | L | P | P | J | J | K | L | P | P | ? | G | W | U | F | S |
| Z | U | Y | N | D | V | W | Q | Z | P | W | H | G | D | J | G | Z | I | T | S | N | X | V | J | H | X | V | D |   | A | M | U |
| X | R | X | H | H | V | N | A | E | N | Q | E | W | F | T | A | E | N | G | I | L | T | T | W | Y | J | Y | Q | H | F | J | O |

The next two letters of known plaintext come from the third X layer and correspond to the case in which the keyed Caesar shift is 4. We will therefore expect the \ configuration to give us KX in the first row in order to be encrypted with SS to give ST.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| S | N | P | I | B | T | P | I | V | K | H | Z | J | T | D | Q | M | T | J | R | M | T | Z | R | X | U | D | P | J | B | N |   |  |
| M | M | H | H | H | Z | K | M | ? | H | R | T | F | J | Z | J | A | R | W | B | H | J | N | S | G | Z | X | P | I | O | I |   |  |
| I | K | U | H | T | J | I | P | V | C | X | X | Q | U | B | B | Z | L | D | S | X | A | J | I | Z | W | U | E | K | N | H | Y |  |

The next region of interest will be positions 65-72 as this will form another complete block of the \ configuration for the already disclosed ERLINCLO plaintext.

Using our lookup table we can see that this will correspond to a right shift of 8 on the underlying cipher text. So in this instance we will be looking for CXUEILMN

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| M | R | Y | Q | K | E | R | A | U | Z | Q | Z | E | K | C | X | Y | D | I | Z | K | U | R | P | T | E | M | S | T | B | E |   |  |
| M | Y | P | K | N | J | N | M | K | V | M | I | Q | K | X | U | A | R | P | E | Z | U | N | H | O | X | Z | Z | I | N | F | U |  |
| N | R | M | D | I | V | C | G | V | R | Z | H | C | V | N | G | L | E | F | N | G | W | J | W | M | D | H | V | R | I | W | I |  |

Rearranging by row shifts to the standard form for the \ configuration gives.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|--|
| C | X | Y | D | I | Z | K | U | R | P | T | E | M | S | T | B | E | M | R | Y | Q | K | E | R | A | U | Z | Q | Z | E | K |   |  |  |
| U | A | R | P | E | Z | U | N | H | O | X | Z | Z | I | N | F | U | M | Y | P | K | N | J | N | M | K | V | M | I | Q | K | X |  |  |
| M | D | I | V | C | G | V | R | Z | H | C | V | N | G | L | E | F | N | G | W | J | W | M | D | H | V | R | I | W | I | N | R |  |  |

As noted at the outset of this paper there is an anomaly here. The expected string for BERLIN would be CXUEILMN, but here we see CXUAILMN instead, which gives us BERLEN rather than BERLIN. The idea that we may have stumbled across an instance of a duress code was mooted earlier, but the author fully takes on board the idea that it may simply mean that this method is incorrect.

Looking for the CK decrypt in the same block but with a keyed Caesar shift of 9 for the / configuration means that we are looking for SK in the order KS, which is duly found as follows:

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| N | Y | P | U | R | F | Y | B | V | K | U | K | F | R | D | Z | P | E | J | K | R | V | Y | T | O | F | N | A | O | C | F |   |  |
| N | P | T | R | Q | L | Q | N | R | W | N | J | U | R | Z | V | B | Y | T | F | K | V | Q | I | S | Z | K | K | J | Q | G | V |  |
| Q | Y | N | E | J | W | D | H | W | Y | K | I | D | W | Q | H | M | F | G | Q | H | X | L | X | N | E | I | W | Y | J | X | J |  |

### X Layer pattern alternation every 32 characters

It will be noticed that we have skipped the B of BERL(I)N in a known section of plaintext. It is reasoned that there is a change of pattern in the order in which the X pattern is read off every 32 characters. The plain text revealed thus far barely touches on this middle section of 32 characters, which is believed to be intentional as it leaves additional hurdles to overcome in the decryption process and further limits the potential attack surface for any algorithmic attempts at brute forcing the result. It would also mean that a three row block similar to those that we have been using as X layers may be involved in some kind of higher level control process as each row has the potential to contain 32 characters. Alternatively this may also have a basis in the 5 bit binary scheme that appears to be hinted at in the layout of K4 whereby rows of 26 are augmented with a 5 character remainder.

Whilst investigating this middle block of 32 characters two alternate schemes were ruled out. First the use of Vigenère decryption as opposed to encryption was attempted as well as an alternate scheme that used left shifts instead of right shifts. Both of these gave inconsistent results. The pattern that satisfies the known plain text and would make sense aesthetically is that the middle block of 32 characters is read as a mirror image of our existing scheme. This means that the rules for the generation of the X layer are the same, but that we are purely changing the order in which the two characters are read off. If we take our original scheme for both configurations as a comparison then we can write out the mirror image for each case.

For the \ configuration.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| U | S | D | J | K | Z | L | D | K | R | N | S | H | G | N | F | I | V | J | E | M | U | F | P | H | Z | L | R | F | A | X | Y |  |
| Y | U | V | L | L | T | R | E | V | J | Y | Q | T | M | K | Y | R | D | M | F | D |   | Y | Q | T | Q | U | X | Q | B | Q | V |  |
| V | F | P | J | U | D | E | E | H | Z | W | E | T | Z | Y | V | G | W | H | K | K | Q | E | T | G | F | Q | J | N | C | E |   |  |

First 32 is read off as: UNYY CKBF.

Alternate 32 is read off as: YYNU FBKC.

Further evidence to back this assertion of a change in the intermediate process to arrive at the correct row alignment is provided further down when re-appraising this scheme in light of the second crib.

### The Second Crib

The rationale for the second crib is based on a fundamental observation about the K3 plain text layout of row 15. I have taken the row that we coloured above to show where the X pattern crosses. The raised letters on the sculpture are shown in a smaller font.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| S | L | O | W | L | Y | D | E | S | P | A | R | A | T | L | Y | S | L | O | W | L | Y | T | H | E | R | E | M | A | I | N | S |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|

The various symmetries that had exposed themselves up to this point made it apparent that a row shift of 16 actually equated to a half rotation over a field of 32. The only exception to the 32 character limit is row 23, so this was thought to be significant. Doing a left or right shift of 16 on this row gives the following result. I shall retain the original configuration directly above.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| S | L | O | W | L | Y | D | E | S | P | A | R | A | T | L | Y | S | L | O | W | L | Y | T | H | E | R | E | M | A | I | N | S |  |
| S | L | O | W | L | Y | T | H | E | R | E | M | A | I | N | S | S | L | O | W | L | Y | D | E | S | P | E | R | A | T | L | Y |  |

It was reasoned from this that the ability to swap these two areas of interest represented something of a kinetic element to the sculpture (with thanks to James Churchill for an explanation of the terminology) that had been hidden using steganography. Whether the metaphor is that of light rays being focused or polarized light being filtered remains to be seen, but further progress was achieved by deciding that this pattern was metaphorical of left and right polarizing filters. Even if the original ANL discovery was only meant to be metaphorical of reflection then it is important to realise that reflections are themselves polarized. I mention this as the next step would appear not to have a basis unless one is working from that principle.

Two points of immediate interest fall out of this proposed piece of steganography that are worth noting. The first is the position of the anomalously raised letters with respect to this virtual overlay as it would seem to indicate that we're on the right track by considering something optical and it may be no coincidence that the corresponding letters in each of the two proposed filters are WL, which may be an intended nod to the words 'wave length', which is of course a major determinant in optical phenomena.

Highlighting the path of the proposed second filter/lens gives this second arrangement.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| S | H | A | D | I | N | G | A | N | D | T | H | E | A | B | S | E | N | C | B | E | T | W | E | E | N | S | U | B | T | L | E |   |
| E | S | T | H | E | N | U | A | N | C | E | O | F | I | Q | L | U | S | I | O | N | * | E | O | F | L | I | G | H | T | L | I |   |
| I | T | W | A | S | T | O | T | A | L | L | Y | I | N | V | I | S | I | B | L | E | H | O | W | S | T | H | A | T | P | O | * |   |
| S | S | I | B | L | E | ? | T | H | E | Y | U | S | E | D | T | H | E | E | A | R | T | H | S | M | A | G | N | E | T | * | * |   |
| I | C | F | I | E | L | D | X | T | H | E | I | N | F | O | R | M | A | T | I | O | N | W | A | S | G | A | T | H | E | R | * |   |
| E | D | A | N | D | T | R | A | N | S | M | I | T | T | E | D | U | N | D | E | R | G | R | U | U | N | D | T | O | A | N | U |   |
| N | K | N | O | W | N | L | O | C | A | T | I | O | N | X | D | O | E | S | L | A | N | G | L | E | Y | K | N | O | W | A | * |   |
| B | O | U | T | T | H | I | S | ? | T | H | E | Y | S | H | O | U | L | D | I | T | S | B | U | R | I | E | D | O | U | T | * |   |
| T | H | E | R | E | S | O | M | E | W | H | E | R | E | X | W | H | O | K | N | O | W | S | T | H | E | E | X | A | C | T | L |   |
| O | C | A | T | I | O | N | ? | O | N | L | Y | W | W | T | H | I | S | W | A | S | H | I | S | L | A | S | T | M | E | S | * |   |
| S | A | G | E | X | T | H | I | R | T | Y | E | I | G | H | T | D | E | G | R | E | E | S | F | I | F | T | Y | S | E | * | * |   |
| V | E | N | M | I | N | U | T | E | S | S | I | X | P | O | I | N | T | F | I | V | E | S | E | C | O | N | D | S | N | O | * |   |
| R | T | H | S | E | V | E | N | T | Y | S | E | V | E | N | D | E | G | R | E | E | S | E | I | G | H | T | M | I | N | U | * |   |
| T | E | S | F | O | R | T | Y | F | O | U | R | S | E | C | O | N | D | S | W | E | S | T | I | D | B | Y | R | O | W | S | * |   |
| S | L | O | W | L | Y | D | E | S | P | A | R | A | T | L | Y | S | L | O | W | L | Y | T | H | E | R | E | M | A | I | N | S |   |
| O | F | P | A | S | S | A | G | E | D | E | B | R | I | S | T | H | A | T | E | N | C | U | M | B | E | R | E | D | T | * | * |   |
| H | E | L | O | W | E | R | P | A | R | T | O | F | T | H | E | D | O | O | R | W | A | Y | W | A | S | R | E | M | O | V | * |   |
| E | D | W | I | T | H | T | R | E | M | B | L | I | N | G | H | A | N | D | S | I | M | A | D | E | A | T | I | N | Y | * | * |   |
| B | R | E | A | C | H | I | N | T | H | E | U | P | P | E | R | L | E | F | T | H | A | N | D | C | O | R | N | E | R | A | N |   |
| D | T | H | E | N | W | I | D | E | N | I | N | G | T | H | E | H | O | L | E | A | L | I | T | T | L | E | I | I | N | * | * |   |
| S | E | R | T | E | D | T | H | E | C | A | N | D | L | E | A | N | D | P | E | E | R | E | D | I | N | T | H | E | H | O | T |   |
| A | I | R | E | S | C | A | P | I | N | G | F | R | O | M | T | H | E | C | H | A | M | B | E | R | C | A | U | S | E | D | * |   |
| T | H | E | F | L | A | M | E | T | O | F | L | I | C | K | E | R | B | U | T | P | R | E | S | E | N | T | L | Y | D | E | T | A |
| I | L | S | O | F | T | H | E | R | O | O | M | W | I | T | H | I | N | E | M | E | R | G | E | D | F | R | O | M | * | * | * |   |
| T | H | E | M | I | S | T | X | C | A | N | Y | O | U | S | E | E | A | N | Y | T | H | I | N | G | Q | ? | O | B | K | R | * |   |
| U | O | X | O | G | H | U | L | B | S | O | L | I | F | B | B | W | F | L | R | V | N | O | R | T | H | E | A | S | T | O | * |   |
| A | W | I | N | F | B | N | Y | P | * | T | W | T | Q | S | J | Q | S | S | E | K | Z | Z | W | A | T | J | K | L | U | D | I |   |
| G | K | U | H | U | A | U | E | K | C | A | R | * | V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J | C | D | I |   |

The second item worthy of note in this arrangement are the two blocks of four characters above and below the WL on the left hand side. Once this region had been highlighted it became obvious that the blocks were respectively two letters short each of the two words **SHADOW FORCES**. The two relevant blocks are shown below. The relevance of this clue will be discussed later when the results of the cryptanalysis are reappraised from this new standpoint.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| S | H | A | D | I | N | G | A | N | D | T | H | E | A | B | S | E | N | C | B | E | T | W | E | E | N | S | U | B | T | L | E |   |  |
| E | S | T | H | E | N | U | A | N | C | E | O | F | I | Q | L | U | S | I | O | N | * | E | O | F | L | I | G | H | T | L | I |   |  |
| I | T | W | A | S | T | O | T | A | L | L | Y | I | N | V | I | S | I | B | L | E | H | O | W | S | T | H | A | T | P | O | * |   |  |
| S | S | I | B | L | E | ? | T | H | E | Y | U | S | E | D | T | H | E | E | A | R | T | H | S | M | A | G | N | E | T | * | * |   |  |
| I | C | F | I | E | L | D | X | T | H | E | I | N | F | O | R | M | A | T | I | O | N | W | A | S | G | A | T | H | E | R | * | * |  |
| E | D | A | N | D | T | R | A | N | S | M | I | T | T | E | D | U | N | D | E | R | G | R | U | U | N | D | T | O | A | N | U |   |  |
| N | K | N | O | W | N | L | O | C | A | T | I | O | N | X | D | O | E | S | L | A | N | G | L | E | Y | K | N | O | W | A | * |   |  |
| B | O | U | T | T | H | I | S | ? | T | H | E | Y | S | H | O | U | L | D | I | T | S | B | U | R | I | E | D | O | U | T | * |   |  |
| T | H | E | R | E | S | O | M | E | W | H | E | R | E | X | W | H | O | K | N | O | W | S | T | H | E | E | X | A | C | T | L |   |  |
| O | C | A | T | I | O | N | ? | O | N | L | Y | W | W | T | H | I | S | W | A | S | H | I | S | L | A | S | T | M | E | S | * |   |  |
| S | A | G | E | X | T | H | I | R | T | Y | E | I | G | H | T | D | E | G | R | E | E | S | F | I | F | T | Y | S | E | * | * |   |  |
| V | E | N | M | I | N | U | T | E | S | S | I | X | P | O | I | N | T | F | I | V | E | S | E | C | O | N | D | S | N | O | * |   |  |
| R | T | H | S | E | V | E | N | T | Y | S | E | V | E | N | D | E | G | R | E | E | S | E | I | G | H | T | M | I | N | U | * |   |  |
| T | E | S | F | O | R | T | Y | F | O | U | R | S | E | C | O | N | D | S | W | E | S | T | I | D | B | Y | R | O | W | S | * |   |  |
| S | L | O | W | L | Y | D | E | S | P | A | R | A | T | L | Y | S | L | O | W | L | Y | T | H | E | R | E | M | A | I | N | S |   |  |
| O | F | P | A | S | S | A | G | E | D | E | B | R | I | S | T | H | A | T | E | N | C | U | M | B | E | R | E | D | T | * | * |   |  |
| H | E | L | O | W | E | R | P | A | R | T | O | F | T | H | E | D | O | O | R | W | A | Y | W | A | S | R | E | M | O | V | * |   |  |
| E | D | W | I | T | H | T | R | E | M | B | L | I | N | G | H | A | N | D | S | I | M | A | D | E | A | T | I | N | Y | * | * |   |  |
| B | R | E | A | C | H | I | N | T | H | E | U | P | P | E | R | L | E | F | T | H | A | N | D | C | O | R | N | E | R | A | N |   |  |
| D | T | H | E | N | W | I | D | E | N | I | N | G | T | H | E | H | O | L | E | A | L | I | T | T | L | E | I | I | N | * | * |   |  |
| S | E | R | T | E | D | T | H | E | C | A | N | D | L | E | A | N | D | P | E | E | R | E | D | I | N | T | H | E | H | O | T |   |  |
| A | I | R | E | S | C | A | P | I | N | G | F | R | O | M | T | H | E | C | H | A | M | B | E | R | C | A | U | S | E | D | * |   |  |
| T | H | E | F | L | A | M | E | T | O | F | L | I | C | K | E | R | B | U | T | P | R | E | S | E | N | T | L | Y | D | E | T | A |  |
| I | L | S | O | F | T | H | E | R | O | O | M | W | I | T | H | I | N | E | M | E | R | G | E | D | F | R | O | M | * | * | * |   |  |
| T | H | E | M | I | S | T | X | C | A | N | Y | O | U | S | E | E | A | N | Y | T | H | I | N | G | Q | ? | O | B | K | R | * |   |  |
| U | O | X | O | G | H | U | L | B | S | O | L | I | F | B | B | W | F | L | R | V | N | O | R | T | H | E | A | S | T | O | * |   |  |
| A | W | I | N | F | B | N | Y | P | * | T | W | T | Q | S | J | Q | S | S | E | K | Z | Z | W | A | T | J | K | L | U | D | I |   |  |
| G | K | U | H | U | A | U | E | K | C | A | R | * | V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J | C | D | I |   |  |

Before arriving at what is deemed to be the second crib arrangement, it is worth highlighting a potential hint already given to us in the last line of K4. The standard arrangement for the first crib puts the double letter block at the right hand side of the last line of K4. If we apply a row shift of 16 to the last row of K4 in standard position then we see CDI is isolated on the left hand side.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| C | D | I | G | K | U | H | U | A | U | E | K | C | A | R | * | V | T | T | M | Z | F | P | K | W | G | D | K | Z | X | T | J |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|

If the ANL component of the CANDIL light is transmitting through the first filter then it seems to be a reasonable assumption that we are expecting the CDI component to be transmitted through the other. Finding a row arrangement scheme capable of representing this was possible under zero shifting of the underlying cipher and plaintext, but a much more aesthetically pleasing result is achieved under a keyed Caesar shift of one, which is what we would naturally expect for the next 8 letters under the proposed deciphering scheme.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| O | X | F | F | Q | A | V | C | O | M | F | A | I | B | E | J | Q | H | B | Q | E | O | I | F | B | C | A | F | Q | D | C | F |
| G | J | U | M | V | A | J | S | Q | * | F | S | G | M | J | H | I | O | M | J | F | A | O | I | F | Q | V | B | Q | D | F | S |
| P | J | Q | W | J | A | J | C | M | F | I | S | X | A | O | I | B | O | T | S | * | J | O | X | B | A | O | S | O | B | M | M |
| A | A | J | C | M | F | ? | O | I | F | P | V | A | F | E | O | I | F | F | B | Y | O | I | A | N | B | H | Q | F | O | * | * |
| J | D | G | J | F | M | E | Z | O | I | F | J | Q | G | S | Y | N | B | O | J | S | Q | X | B | A | H | B | O | I | F | Y | * |
| F | E | B | Q | E | O | Y | B | Q | A | N | J | O | F | E | V | Q | E | F | Y | H | Y | V | V | Q | E | O | S | B | Q | V | * |
| Q | R | S | X | Q | M | S | D | B | O | J | S | Q | Z | E | S | F | A | M | B | Q | H | M | F | P | R | Q | S | X | B | * |   |
| C | S | V | O | O | I | J | A | ? | O | I | F | P | A | I | S | V | M | E | J | O | A | C | V | Y | J | F | E | S | V | O | * |
| O | I | F | Y | F | A | S | N | F | X | I | F | Y | F | Z | X | I | S | R | Q | S | X | A | O | I | F | F | Z | B | D | O | M |
| S | D | B | O | J | S | Q | ? | S | Q | M | P | X | X | O | I | J | A | X | B | A | I | J | A | M | B | A | O | N | F | A | * |
| A | B | H | F | Z | O | I | J | Y | O | P | F | J | H | I | O | E | F | H | Y | F | F | A | G | J | G | O | P | A | F | * | * |
| W | F | Q | N | J | Q | V | O | F | A | A | J | Z | T | S | J | Q | O | G | J | W | F | A | F | D | S | Q | E | A | Q | S | * |
| Y | O | I | A | F | W | F | Q | O | P | A | F | W | F | Q | E | F | H | Y | F | F | A | F | J | H | I | O | N | J | Q | V | * |
| O | F | A | G | S | Y | O | P | G | S | V | Y | A | F | D | S | Q | E | A | X | F | A | O | Z | M | B | P | F | Y | O | X | S |
| A | M | S | X | M | P | E | F | A | T | B | Y | B | O | M | P | A | M | S | X | M | P | O | I | F | Y | F | N | B | J | Q | A |
| S | G | T | B | A | A | B | H | F | E | F | C | Y | J | A | O | I | B | O | F | Q | D | V | N | C | F | Y | F | E | O | * | * |
| I | F | M | S | X | F | Y | T | B | Y | O | S | G | O | I | F | E | S | S | Y | X | B | P | X | B | A | Y | F | N | S | W | * |
| F | E | X | J | O | I | O | Y | F | N | C | M | J | H | I | B | Q | E | A | J | N | B | E | F | B | O | J | Q | P | * | * |   |
| C | Y | F | B | D | I | J | Q | O | I | F | V | T | T | F | Y | M | F | G | O | I | B | Q | E | D | S | Y | Q | F | Y | B | Q |
| E | O | I | F | Q | X | J | E | F | Q | J | H | O | I | F | I | S | M | F | B | M | J | O | O | M | F | J | J | Q | * | * |   |
| A | F | Y | O | F | E | O | I | F | D | B | Q | E | M | F | B | Q | E | T | F | F | Y | F | E | J | Q | O | I | F | I | S | O |
| B | J | Y | F | A | D | B | T | J | Q | H | G | Y | S | N | O | I | F | D | I | B | N | C | F | Y | D | B | V | A | F | E | * |
| O | I | F | G | M | B | N | F | O | S | G | M | J | D | R | F | Y | C | V | O | T | Y | F | A | F | Q | O | M | P | E | F | O |
| J | M | A | S | G | O | I | F | Y | S | S | N | X | J | O | I | J | Q | F | N | F | Y | H | F | E | G | Y | S | N | * | * |   |
| O | I | F | N | J | A | O | Z | D | B | Q | P | S | V | A | F | F | B | Q | P | O | I | J | Q | H | U | ? | S | C | R | Y | * |
| U | T | Y | Q | H | R | A | A | S | * | V | S | Z | S | H | I | V | M | C | A | S | M | J | G | C | C | X | G | M | Y | W |   |
| R | K | K | X | B | O | L | R | M | V | E | J | B | X | J | Q | G | C | Q | P | T | * | O | X | O | U | A | L | U | A | A | F |
| Y | * | W | O | O | N | K | G | T | R | X | H | E | R | K | Z | O | L | D | E | J | H | R | V | I | V | B | V | F | R | D | B |
| V | G | T | I | K | M | Y | G | B | Z | P | V | A | E | L | R | K | M | E | R | Y | Q | A | I | H | Q | G | J | W | L | F | N |
| O | N | R | P | Y | E | N | G | E | * | P | U | O | U | V | Z | U | C | U | W | P | V | W | M | M | O | Y | F | W | L | P | U |
| F | O | K | P | W | H | X | I | R | R | U | F | O | H | G | U | L | Q | D | F | * | W | G | T | L | V | E | F | F | I | K | X |
| H | H | X | I | R | R | ? | E | U | N | D | T | G | U | K | E | U | N | N | J | B | H | T | G | Z | I | U | Y | M | H | * | * |
| O | J | N | W | N | K | L | B | Q | U | M | W | R | U | F | E | B | H | E | W | G | Y | T | L | V | Q | H | F | V | Q | B | * |
| U | K | H | K | M | F | D | H | P | V | Z | V | F | F | O | L | O | C | L | M | C | U | D | Y | O | C | L | E | G | I | Y | Y |
| P | J | K | F | O | R | K | F | N | W | E | V | G | R | A | L | I | R | G | X | I | R | V | X | U | M | A | K | G | O | J | * |
| I | I | E | E | E | V | W | I | ? | E | X | R | C | G | V | G | T | X | Q | O | E | G | J | P | D | V | U | K | F | Y | F | * |
| F | W | M | E | R | G | F | K | N | S | U | U | L | M | O | O | V | H | A | P | U | T | G | F | V | Q | M | B | W | J | E | Z |
| G | M | H | H | O | F | K | ? | G | R | K | C | A | G | E | U | W | H | S | H | J | T | V | G | Z | I | E | Y | R | G | * |   |
| G | I | U | Q | O | H | T | V | B | F | D | Q | V | W | T | E | L | N | U | D | M | U | V | N | V | Q | F | E | G | U | * | * |
| F | M | K | K | W | Y | T | H | R | G | G | W | S | F | F | Z | C | E | N | W | T | O | G | U | Z | F | K | M | H | Y | F | * |
| E | Q | U | G | N | T | Q | K | H | M | G | M | T | N | Y | L | U | P | B | M | N | H | Q | V | W | T | E | Z | W | R | T | * |
| E | U | V | N | F | C | F | E | N | I | E | B | G | N | L | H | K | Q | V | T | M | H | F | A | X | L | M | M | B | F | O | H |
| F | Q | E | P | B | I | Y | S | I | Q | M | A | Y | I | F | S | D | T | O | F | S | J | C | J | E | P | A | I | Q | B | J | B |
| D | I | O | Q | Y | F | P | V | M | E | A | M | M | A | M | M | Q | S | I | A | Q | S | A | N | Y | X | Z | N | Q | F | * | * |
| O | T | Y | Q | H | B | O | J | I | Q | Y | B | Y | T | F | A | M | Q | Q | F | M | F | C | M | T | J | J | B | D | B | F | * |
| X | N | O | X | Q | E | J | O | F | F | Q | Y | B | I | D | O | F | Q | F | V | E | Y | F | O | Q | I | B | F | S | F | * | * |
| O | G | S | M | A | F | E | O | J | X | F | Q | I | B | F | J | S | P | O | F | P | U | I | F | F | Q | D | O | B | P | D | Y |
| F | J | G | O | C | Y | A | T | B | N | I | I | F | X | F | Q | B | O | B | N | B | O | F | H | P | F | F | Y | M | C | * | * |
| O | F | F | G | S | B | A | G | J | S | O | V | F | O | V | B | F | S | O | S | B | Y | N | B | F | F | Y | O | Q | Y | O | J |
| C | A | F | E | E | Q | J | B | B | I | O | O | N | A | O | F | X | T | J | F | Y | S | B | H | Y | J | F | X | G | F | C | * |
| B | F | D | O | E | E | I | J | M | D | F | J | I | A | J | O | F | H | S | F | B | S | A | E | E | Y | P | E | M | S | Y | J |
| Y | R | M | N | M | F | I | B | H | O | E | I | B | Y | E | T | O | F | S | I | N | H | G | N | G | F | V | I | F | * | * | * |
| F | D | E | N | Y | J | T | G | F | J | N | F | I | Q | M | A | A | O | O | Y | O | W | E | S | I | X | ? | S | C | R | Y | * |
| V | S | Z | S | H | I | V | M | C | A | S | M | J | G | C | C | X | G | M | Y | W | U | T | Y | Q | H | R | A | A | S | * |   |
| O | X | O | U | A | L | U | A | A | F | R | K | K | X | B | O | L | R | M | V | E | J | B | X | J | Q | G | C | Q | P | T | * |
| W | O | O | N | K | G | T | R | X | H | E | R | K | Z | O | L | D | E | J | H | R | V | I | V | B | V | F | R | D | B | Y | * |

There are a few observations that we can initially make, the first is that where we initially had ANL in the light path of the first filter we have retained an equivalent NLA. We have used the same row shift rules as applied previously in that K4 and K1/2 rows are being moved in opposition to each other. As a result we can see that the light path for DCI is not DCI but OHQ. Usage of the term OHB (old headquarters building) appears to be occasionally interchangeable with OHQ, so this was deemed to be significant given that the ground breaking ceremony of the OHQ and Kryptos both occurred on the same day separated by some 31 years.

Prior to the discovery of the row alignment process one of the most intriguing strings to drop out of the allowed form for the / configuration under a shift of 1 was GWDI PLZW as shown in this zoomed screenshot of the software.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| I | H | Q | G | J | W | L | F | N | V | G | T | I | K | M | Y | G | B | Z | P | V | A | E | L | R | K | M | E | R | Y | Q | A |
| W | L | P | U | O | N | R | P | Y | E | N | G | E | * | P | U | O | U | V | Z | U | C | U | W | P | V | W | M | M | O | Y | F |
| I | R | R | U | F | O | H | G | U | L | Q | D | F | * | W | G | T | L | V | E | F | F | I | K | X | F | O | K | P | W | H | X |

This string encrypts with GHULBSOL to give KEDREWTH. Such a string in isolation looks odd, but under the decryption of the first 8 letters found in the first crib this would have given us:

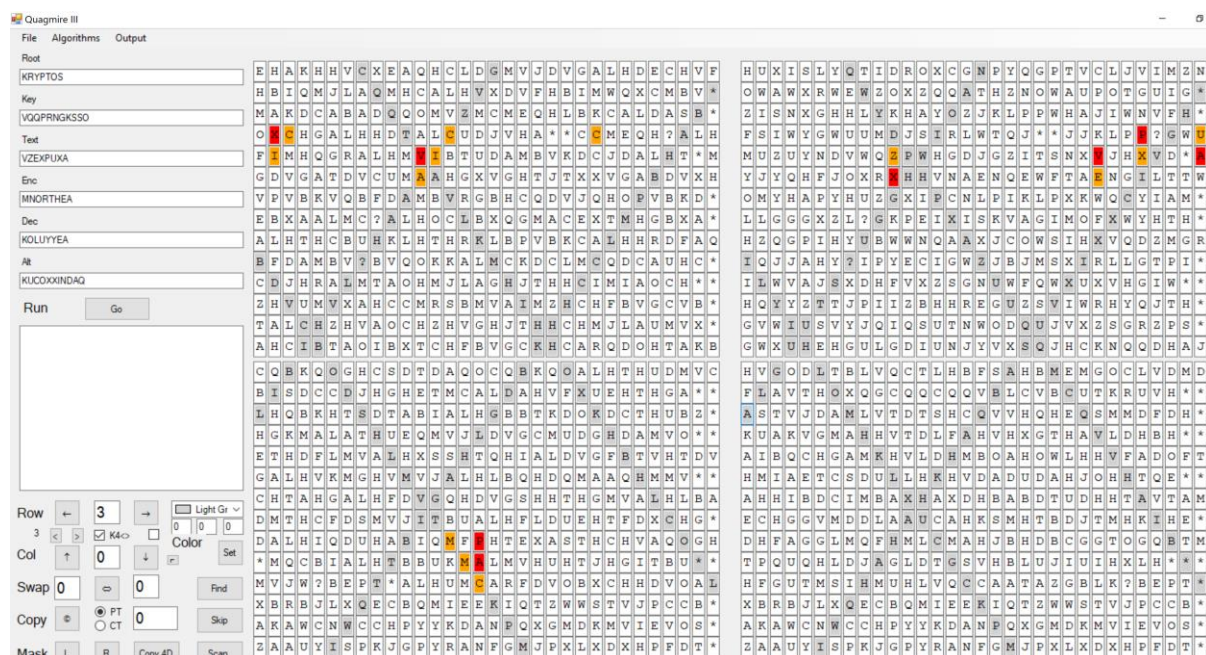
KRYPTOS IKE DREW TH

I draw on this as an example as this a case of an allowable string that is generated by the rules that our previous cryptanalysis has sketched out based on the known plaintext, but has no other supporting keys or underlying rationale other than the coincidence that it occurs under a keyed Caesar shift of 1 and is of the correct form for the / configuration. Despite these tighter constraints there are still over 20,000 potential solutions for each shift setting when trying to arrive at the next

block of 8, so we obviously need to apply other constraints. The first of these was found in the first crib when we found that the UY and CD of our target blocks were found embedded in our row alignment scheme. Applying this principle then it should be possible to work backwards to see if we can find pointers towards any tighter constraints when re-appraising the known plain text in the light of our current decryption model.

## Revisiting the plain text

Let's examine the NORTH WEST clue again first. We know according to our rules that this is the / configuration and that the bottom row can only be in a position where there is a Z directly above an X. This immediately limits the potential positions of our proposed row alignment. Arranging for this position with the double filter highlighting on shows the following relationship. Notice that we are also using the symmetrically opposed 3 bottom rows within the alignment process as well as was done with the first crib.



In the plaintext (under a right shift of 3 in the keyed Caesar scheme) the left hand filter can be seen to have CIA running the course of the 3 rows in the 'beam' position. If the block XC (highlighted on the left) was above IB then this would be a perfect mirroring of the example in crib 1, where XV would form a diagonal against the CIA string that falls along the 'beam'.

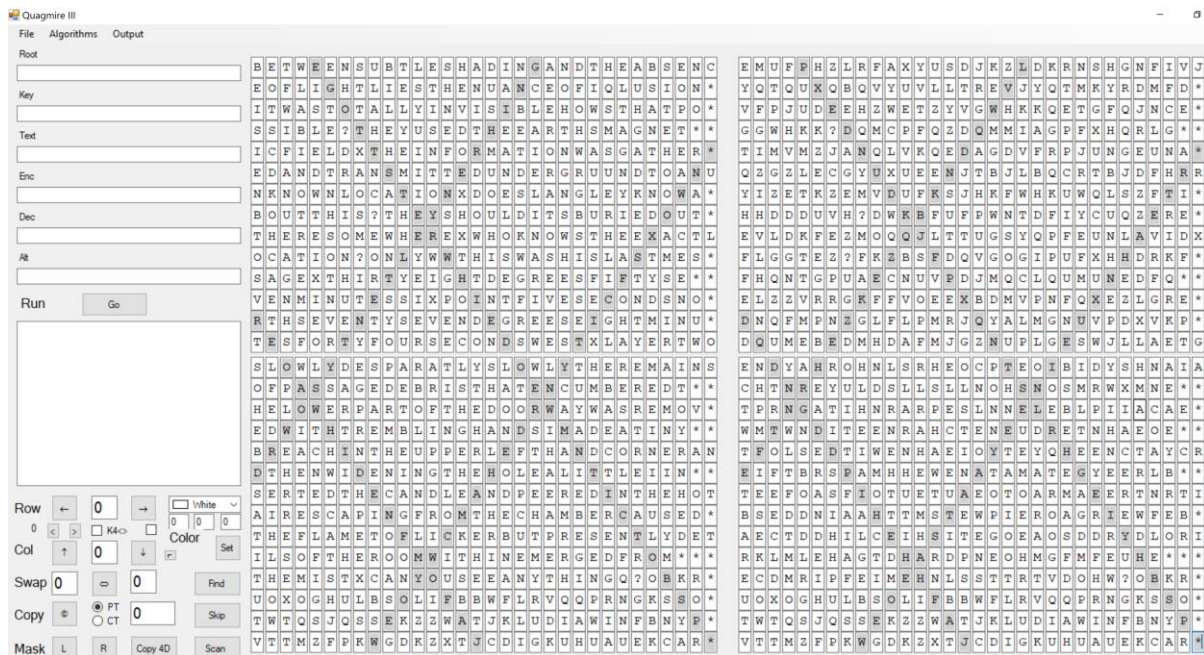
But something else appears to be in play here and it's presently not clear what is being missed. In the first crib we saw that double Y position was interchangeable, due to 3 Y's being in the 2<sup>nd</sup> row each separated by 9 letters, which of course led to our initial break. In this case the C's in the first row have a similar relationship, although in this instance they are separated by 10 characters. If the first row was right shifted 10 characters then the correct diagonal VX marker would be formed to complete the NORTH EAST clue, but this would actually destroy the very pattern we are looking for as the corresponding cipher text row would also be shifted by 10 characters. So here is a case in point of running into a situation through reverse engineering where we have not done the initial work required in the previous steps in order to make sense of the situation which we are observing.

It was mentioned during the cryptanalysis section that it is believed that this method of determining the row alignments changes every 32 characters, but that the guiding form of the proposed decryption scheme remains unaltered. To illustrate this point we now revisit the BERL(I)N clue

armed with our knowledge of the first crib and our previous treatment of the NORTH EAST example. The position of the BERL(I)N clue is incredibly interesting as it occurs at the point of cross over between the top and bottom panels of the sculpture. Remember that the marker scheme needs to identify both the left and right participating blocks. Here we see direct evidence that there is a change of scheme in how the markers are placed as we actually find them embedded this time round in the cipher text itself, but in the correct positions despite the rather cramped circumstances. Remember that the scheme would usually expect to be 3 lines for both the top and bottom sections, but due to the crossover it will be seen that this is compressed into a space of four rows, giving this rather elegant relationship. It was also be obvious once again that both the filter position and the letters CIA are significant in the row alignment scheme. Remember that this arrangement assumes that gives a decryption of BERLEN rather than BERLIN as previously discussed in the cryptanalysis section above. Remember that we are dealing with the \ configuration with a shift of 8 for this block of 8.



We can see that for the next block selection we will actually be completely within the bottom left hand panel of the sculpture for our letter selection process. It is reasoned that this is the purpose of the SHADOW FORCES clue. By specifying the positions of these two blocks it reinforces the idea the letters WL represent either a focal point or filter boundary. Therefore as we make the crossover from the top to the bottom panel it is reasoned that we need to redraw our two filters as follows.



## Further Research

Further research is now focused on refinement of the first and second crib examples as many more phenomena not published in this paper show different results can be obtained by shifting the bottom rows left or right with respect to row movements in the top half. By further experimentation with these patterns it is reasoned that breaking the second crib for the / configuration will finally show the logical step we are missing in our decryption of the NORTH EAST clue as NORTHEA occurs within the same block of 32 characters. It is believed that further progress will only be made by understanding and following the specific path laid out by the puzzles creator in order, but it is hoped that this paper inches us closer to understanding the cryptic, but logical language that has been used in its construction.